Jeetendra R Alur

Computational Physics · Scientific ML · Nonlinear Dynamics

Bengaluru, India | jeetendrar007@gmail.com | +91-8095575286 | Website | github.com/Jeetendraralur

Research Interests

Scientific Machine Learning · Nonlinear Dynamics · PINNs · Chaos theory · Hamiltonian Systems

Education

BSc Honours Physics with Research, 2022 – 2026

Deen Dayal Upadhayay College, Delhi University,

- Thesis: *"Hamiltonian Physics-Informed Neural Networks for Chaotic Double Pendulum Dynamics"*, supervised by Prof. Kulvinder Singh
- Relevant Coursework: Classical Mechanics, Quantum Mechanics, Statistical Mechanics, Mathematical Physics (I–III), Group Theory and Representations, Digital Electronics, Nuclear and Particle Physics, Research Methodology

BS in Data Science and Applications 2023 – present

IIT Madras (in-progress)

- Relevant Coursework: Statistics, Data Structures and Algorithms, Machine Learning Foundations, Mathematics

Research Experience

Thesis research 2025 – 2026

DU, Prof. Kulvinder Singh

- Derived Hamiltonian formulation of a double pendulum and implemented physics informed neural network
- Benchmarked H-PINN predictions with respect to RK4 integrator and quantified the results, the prediction horizons, using Largest Lyapunov Exopnent λ
- Enforced energy conservation by constraining overtime energy loss and improve long term consistency
- Investigated active control by adding counter torque to the system

Selected Projects

Machine Learning in Physics 2023

Course/Project - Hindu College

- Built a PINN in PyTorch to solve Kuramoto–Sivashinsky equation using automatic differentiation

Fourier Optics and Computational Imaging 2024

Course/Project - Hindu College

- Simulated diffraction, Fourier transforms, spatial filtering, and 4F optical systems
- Developed an edge detection and real-time image processing application using multiple filtering techniques as the final project.
- Link: https://github.com/Jeetendraralur/edge-detection-fourier-optics-and-computational-imaging-

Technical Skills

Languages	Python, MATLAB, LaTeX
Computational	PyTorch, NumPy/SciPy, scikit-learn, Matplotlib, Mathematica
Lab / Instrumentation	Experimental Physics laboratory skills, electronic instrumentation and analysis